

Solana SVM Study Course

Course Overview

Course Title

Mastering Solana and SVM Development: From EVM to Solana

Duration & Level

- **Duration:** 16 weeks (part-time) / 8 weeks (full-time)
- **Level:** Intermediate to Advanced
- **Prerequisites:** JavaScript/TypeScript, basic blockchain concepts, familiarity with EVM development

Learning Objectives

- Master Solana blockchain architecture and SVM (Solana Virtual Machine)
- Build Solana applications using NestJS
- Understand key differences between EVM and SVM paradigms
- Implement blockchain solutions
- Deploy and monitor Solana applications in production

Course Repository Structure

```
solana-svm-study/
├── docs/
│   ├── COURSE.md
│   ├── STUDY.md
│   ├── TASKS.md
│   ├── MASTER-ITERATION.md
│   └── diagrams/
├── src/
│   ├── modules/
│   │   ├── accounts/
│   │   ├── transactions/
│   │   ├── tokens/
│   │   └── ...
│   ├── common/
│   ├── database/
│   └── shared/
├── infra/
├── scripts/
└── ...
```

Course documentation
This course curriculum
Detailed study topics
Implementation tasks
Project philosophy
Architecture diagrams
Application source code
Feature modules
Account management
Transaction services
SPL token operations
Additional modules
Shared utilities
Database layer
Shared types
Infrastructure as code
Utility scripts
Configuration files

Technology Stack

Backend Framework

- **NestJS:** Progressive Node.js framework
- **TypeScript:** Type-safe JavaScript
- **TypeORM:** TypeScript ORM for databases

Blockchain Integration

- **Solana Web3.js:** Solana blockchain interaction
- **@solana/web3.js:** Official Solana JavaScript SDK

Database & Messaging

- **PostgreSQL:** Primary database
- **Apache Kafka:** Event streaming

Development Environment

Local Development

- Docker Compose for services
- Hot reload with NestJS CLI
- Integrated testing with Jest
- Code coverage reporting

Production Deployment

- Kubernetes manifests
- CI/CD with GitHub Actions
- Multi-stage Docker builds
- Health checks and monitoring

Course Modules Overview

Core Modules

1. **Accounts & Programs** - Account management and PDAs
2. **Transactions & Instructions** - Transaction building and submission
3. **Token Standards** - SPL token operations
4. **Account Abstraction** - Smart accounts and PDAs
5. **Fee Mechanism** - Transaction fees and prioritization

Advanced Modules

6. **Consensus & Validation** - Block validation and consensus
7. **Signing & Cryptography** - Digital signatures and keys
8. **MPC** - Multi-party computation protocols
9. **SVM** - Solana Virtual Machine integration

Learning Approach

Hands-on Implementation

- Build complete NestJS API for Solana integration
- Implement all major SVM features
- Real-world blockchain application development

Progressive Complexity

- Start with basic concepts (accounts, transactions)
- Build up to advanced features (MPC, CPIs)
- End with production deployment and monitoring

EVM to SVM Migration

- Direct comparisons between EVM and SVM concepts

Assessment & Evaluation

Implementation Tasks

- Complete all module implementations
- Pass comprehensive test suites
- Code review and optimization

Project Milestones

- Database schema and migrations
- API endpoint implementation
- Integration testing
- Performance optimization

Final Deliverables

Success Metrics

Technical Proficiency

- 80%+ test coverage
- Performance benchmarks met
- Security best practices implemented
- Production deployment successful

Code Quality

- TypeScript strict mode compliance
- Clean architecture patterns
- Comprehensive error handling
- API documentation complete

Thank You!

Questions & Next Steps

- Review the [STUDY.md](#) for detailed topics
- Check [TASKS.md](#) for implementation guide
- Start with [Module 1: Accounts & Programs](#)

Happy Learning! 🚀